Anjuman-i-Islam's

M. H. Saboo Siddik College of Engineering

8, Saboo Siddik Polytechnic Road, Byculla, Mumbai-8

Internal Assessment Examination –I (IAE-I)

February 2026

Blue Print for Students Information

Course Code:	2114112	Course Name: Database Management System	Sem: IV

Table 1: Course objectives and Outcomes

Course Objectives	
CO1	To Understand the fundamentals of a database systems
CO2	Develop entity relationship data model /EER and its mapping to relational model
CO3	Learn relational algebra and Formulate SQL queries.
CO4	Apply normalization techniques to normalize the database
CO5	Understand concept of transaction, concurrency control and recovery techniques
CO6	Explore and understand recent databases and their application

2164112	Course Outcomes
2164112.1	Understand concepts of DBMS and design ER/EER diagrams for real-world applications.
2164112.2	Apply mapping rules to construct relational models and formulate relational algebra queries.
2164112.3	Apply SQL queries for database definition, manipulation, and control operations.
2164112.4	Analyze and apply normalization techniques to relational database design.
2164112.5	Understand transaction management, concurrency control, and recovery techniques.
2164112.6	Understand recent database technologies such as NoSQL, cloud, and distributed databases.

Table 2: Detail Syllabus Modules for IAE-I

Module No.	Content/ Sub Modules	CO	No of Hrs in Syllabus	Marks Weightage	% of Syllabus	Marks Range (Including Options) [20-32]
1	Introduction to Database and Data Modeling	1	8	47%	21	9-14
	Introduction: Definitions and application, Characteristics of databases, DBMS architecture, ACID Properties The Entity-Relationship (ER) Model: Entity types: Weak and strong entity sets, Entity sets, Types of Attributes, Keys, Relationship constraints: Cardinality and Participation, Extended Entity-Relationship (EER) Model: Generalization, Specialization and Aggregation					

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2	Relational Model and Relational Algebra	2	5	29%	13	6-9
	Introduction to the Relational Model, relational schema and concept of keys. Mapping the ER and EER Model to the Relational Model, Relational Algebra-operators (Selection(σ),Projection(π),Union(∪),Difference (−),CartesianProduct(×),Join(⋈), Intersection (∩),Rename (ρ)), Relational Algebra Queries					
3	Structured Query Language	3	5	24%	10	5-9
	Overview of SQL, Data Definition Commands, Integrity constraints: key constraints, Domain Constraints, Referential integrity, check constraints, Data Manipulation commands, Data Control commands					

Dr. Muhib Anwar Lambay
Course faculty

Er.Chaitali Mahajan
Department Exam Co-Ordinator

Dr. Irfan Landge
HOD (CSE AI & ML)